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Degree: B.Tech

Roll no.:.....
Semester: 3

Mid Semester Examination, September 2025

Course Code: ITMTC301/INMTC301

Course title: Optimization Principles and Techniques

Time: 1 hour 30 minutes

Maximum Marks. 20

Note: Attempt all the questions in the given order only. Missing data/information(if any), may be suitably assumed and mentioned in the answer.

Q. No.	Questions	Marks	CO
1a	Solve the following LPP using graphical method: Max $Z = 2x_1 + 3x_2$ subject to $x_1 + x_2 \leq 30,$ $x_1 - x_2 \geq 0,$ $x_2 \geq 3,$ $0 \leq x_1 \leq 20,$ $0 \leq x_2 \leq 12.$	2	CO1
1b	Solve the following linear programming problem. Max $Z = x + 3y + 6z - w$ subject to $5x + y + 6z + 7w \leq 20,$ $6x + 2y + 2z + 9w \geq 40,$ $x, y, z, w \geq 0.$	2	CO1
2a	Find all degenerate basic feasible solutions of the system $x_1 + x_2 + x_3 = 3,$ $x_1 - x_2 + x_4 = 0,$ $x_1, x_2, x_3, x_4 \geq 0.$	2	CO1
2b	A retired person plans to invest up to Rs. 50,000 in NSC certificates yielding 12 percent interest per year and corporate bonds yielding 15 percent interest per year. The person is told to invest a minimum of Rs. 25,000 in the NSC certificates and a minimum of Rs. 10,000 in the corporate bonds, with no more than one-fourth of the total investment to be in corporate bonds. Formulate the given problem as an LPP to decide how the manager should invest in each type of bond to achieve the maximum annual interest.	2	CO1
3a	Solve the following LPP using simplex method and discuss the nature of the solution: Min $Z = -4x_1 + x_2$ subject to $x_1 - 2x_2 \leq 2,$ $-2x_1 + x_2 \leq 2,$ $x_1, x_2 \geq 0.$	2	CO1
3b	Consider a balanced transportation problem (TP) having 2 sources and 3 destinations such that the availability at each source is 30 units and the demand at each destination is 20 units. Given that the dual variables in the optimal table corresponding to sources and destinations are -1, 2, and 1, 2, 3, respectively. Then, find the optimal objective function value of the given TP.	2	CO2

4a	Consider the LPP Max $6x_1 + 12x_2 + 12x_3 - 6x_4$ subject to $x_1 + x_2 + x_3 = 4,$ $x_1 + 4x_2 + x_4 = 8,$ $x_1, x_2, x_3, x_4 \geq 0.$ The optimum table is given by: <table><tr><th>Basis</th><th>Solution</th><th>x_1</th><th>x_2</th><th>x_3</th><th>x_4</th></tr><tr><td>x_3</td><td>2</td><td>3/4</td><td>0</td><td>1</td><td>-1/4</td></tr><tr><td>x_2</td><td>2</td><td>1/4</td><td>1</td><td>0</td><td>1/4</td></tr><tr><td>z</td><td>48</td><td>6</td><td>0</td><td>0</td><td>6</td></tr></table> Write the dual problem and find the solution of the dual problem from the optimal solution of primal problem without solving the dual.	Basis	Solution	x_1	x_2	x_3	x_4	x_3	2	3/4	0	1	-1/4	x_2	2	1/4	1	0	1/4	z	48	6	0	0	6	2	CO2	
Basis	Solution	x_1	x_2	x_3	x_4																							
x_3	2	3/4	0	1	-1/4																							
x_2	2	1/4	1	0	1/4																							
z	48	6	0	0	6																							
4b	Solve the following LPP using dual simplex method: Min $Z = x_1 + x_2$ subject to $2x_1 + x_2 \geq 4,$ $x_1 + 7x_2 \geq 7,$ $x_1, x_2 \geq 0.$	2	CO2																									
5a	Justify the following statements with proper reasoning whether these are correct or incorrect. 1. If primal has no feasible solution then dual has always unbounded solution. 2. A primal and dual both can have an unbounded optimal solution.	2	CO2																									
5b	Consider the problem of assigning 4 operators O_1, O_2, O_3, O_4 to 4 machines M_1, M_2, M_3, M_4 . The assignment cost (in Rs.) is given in the following table. <table><tr><td></td><td>M_1</td><td>M_2</td><td>M_3</td><td>M_4</td></tr><tr><td>O_1</td><td>18</td><td>26</td><td>17</td><td>11</td></tr><tr><td>O_2</td><td>13</td><td>28</td><td>14</td><td>26</td></tr><tr><td>O_3</td><td>38</td><td>19</td><td>18</td><td>15</td></tr><tr><td>O_4</td><td>19</td><td>26</td><td>24</td><td>10</td></tr></table>		M_1	M_2	M_3	M_4	O_1	18	26	17	11	O_2	13	28	14	26	O_3	38	19	18	15	O_4	19	26	24	10	2	CO2
	M_1	M_2	M_3	M_4																								
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